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DAY 8

COMPUTER VISION

How AI Sees the World — From Phone Cameras to Self-Driving Cars

A Course Guide by AYE Tech Hub

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1. Introduction

Sight is the most information-rich of our senses, and giving machines the ability to see has been one of the great pursuits of AI since the 1960s. After more than fifty years of slow progress, the deep-learning revolution of 2012 changed everything almost overnight. Today, **computer vision** powers face unlock on your phone, defect inspection in the factories of Gimpo and Mekelle, the cameras of self-driving Tesla and Hyundai cars, medical AI that spots tumours before doctors can, and Google Photos that lets you search for 'beach' or 'dog' across thousands of images.

Computer vision is also one of the most practical AI fields for a working engineer. Unlike language models which run in the cloud and cost money per query, vision models often run on small, cheap hardware right next to the camera — at the **edge** — and can be deployed on a single industrial PC. If you understand how vision models work, you can build inspection systems, quality-control stations, and safety monitors yourself.

Learning Objectives

By the end of this lesson you will be able to:
- Explain the six-stage computer vision pipeline from capture to decision.
- Distinguish the four core vision tasks: classification, detection, segmentation, pose.
- Describe how a CNN builds a hierarchy of features from pixels to objects.
- Compare CNNs and Vision Transformers (ViT) and pick the right one for a task.
- Outline how factory visual inspection systems are designed end-to-end.
- Decide when a vision model should run on the edge versus in the cloud.

2. The Vision Pipeline

Every computer vision system — from a $20 webcam to a $250 000 medical scanner — follows the same six-stage pipeline. Understanding each stage helps you debug what is going wrong when a system performs poorly. In real installations, the AI model itself (stages 3 and 4) often gets all the attention — but it is the unglamorous stages (capture, preprocessing, post-processing) that decide whether the system works in production.

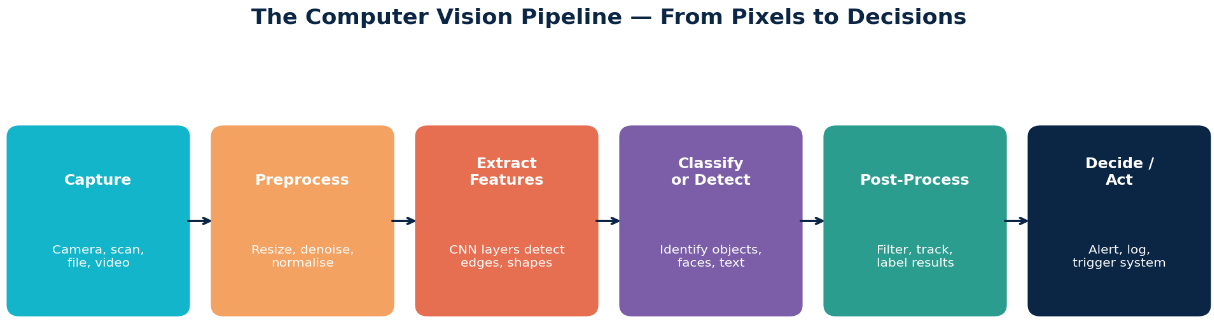

Figure 2.1 — The six-stage computer vision pipeline. Each stage transforms the data: capture → preprocess → extract features → classify → post-process → decide.

Stage	What Happens	Common Failure Points
1. Capture	Camera, scan, file, video stream	Bad lighting, motion blur, lens dirt
2. Preprocess	Resize, denoise, normalise brightness	Wrong colour space, lost detail
3. Extract features	CNN or ViT layers find edges, shapes, parts	Model trained on different data
4. Classify / detect	Identify what is in the image	Confused by lookalike objects
5. Post-process	Filter false positives, track over time	Threshold too strict or too loose
6. Decide / act	Alert, log, trigger PLC or system	Wrong action taken, no fail-safe

3. Four Core Vision Tasks

Almost every real-world vision application breaks down into one of four task types. Each requires a different model architecture, a different way of labelling training data, and a different evaluation metric. The first decision in any vision project is choosing which task you are solving — getting that wrong wastes weeks of work.

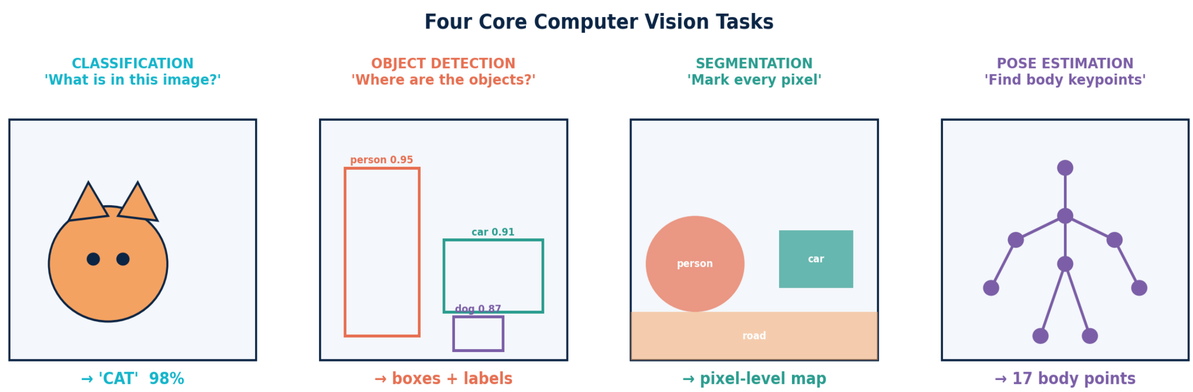

Figure 3.1 — The four core computer vision tasks. From simplest (classification) to most complex (pose estimation). Each requires more detailed training data than the last.

Task	Output	Example Use	Common Tools
Classification	One label for the whole image	'Is this a defective part?'	ResNet, EfficientNet, ViT
Object Detection	Bounding boxes + labels	'Where are the cars in this image?'	YOLO, RT-DETR, Faster R-CNN

Task	Output	Example Use	Common Tools
Segmentation	Pixel-level mask per object	'Exactly which pixels are the tumour?'	U-Net, Mask R-CNN, SAM
Pose Estimation	Skeleton keypoints	'What is this person's body pose?'	OpenPose, MediaPipe, MMPose

Engineer's Tip

Start with the simplest task that solves your problem. If you only need to know whether a part is defective (yes/no), use **classification** — not detection. Simpler tasks need much less labelled data (1000 examples vs 10 000) and run faster on small hardware. Upgrade to detection or segmentation only when classification cannot answer the question.

4. CNNs — How Vision Networks Build Understanding

Convolutional Neural Networks (CNNs) remain the workhorse of computer vision in 2026. Their genius is the **feature hierarchy**: each layer takes the output of the previous layer and combines it to detect more abstract patterns. The first layers see only edges and colour blobs; middle layers see textures and shapes; deep layers see whole objects. The same architecture that recognises cats also recognises welds, tumours, traffic signs, and circuit board defects — only the training data changes.

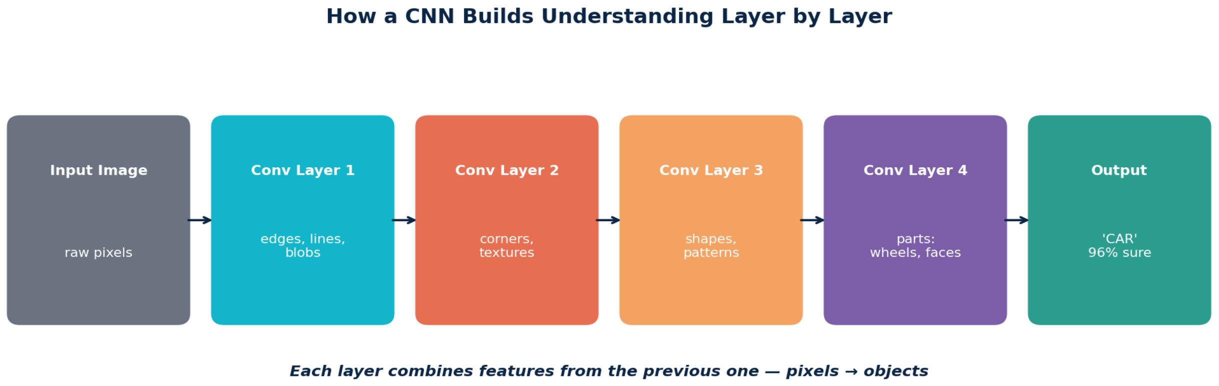

Figure 4.1 — How a CNN builds increasingly abstract features from raw pixels. Each layer combines simpler features from below to detect higher-level concepts.

Why CNNs Work So Well on Images

Three design choices make CNNs perfect for images: (1) **local connectivity** — each neuron looks at a small patch of the image, not the whole thing, so the network has far fewer parameters; (2) **weight sharing** — the same filter slides across the entire image, so a 'cat ear detector' learned in the top-left works equally well in the bottom-right; (3) **pooling** — gradually reducing image size as the network goes deeper, which makes the model robust to small position shifts. These three ideas combined make CNNs 100× more efficient than fully-connected networks on images.

5. CNN vs Vision Transformer (ViT)

Since 2020, a new architecture has emerged for vision: the **Vision Transformer (ViT)**. Instead of sliding filters across the image, ViT splits the image into small patches (e.g., 16×16 pixels), treats each patch like a 'token', and applies the same self-attention mechanism you learned about in Day 7. At very large scale, ViTs outperform CNNs — but CNNs remain the practical choice for most engineering work because they are faster, cheaper, and need less data.

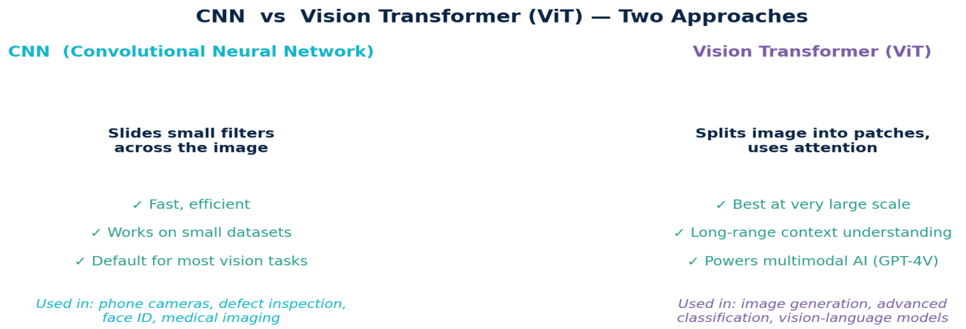

Figure 5.1 — CNNs and Vision Transformers solve the same problems differently. CNNs are the default for most engineering work; ViTs dominate at the cutting edge of multimodal AI.

6. Factory Visual Inspection

One of the most valuable engineering applications of computer vision is **automated quality inspection**. A camera mounted above a conveyor captures every part; a CNN running on a small industrial PC classifies each one as OK or defective in 30 ms; a PLC fires a reject arm to push bad parts off the line. The same setup — with different training data — works for bottle caps, circuit boards, welded joints, automotive parts, or pharmaceutical pills.

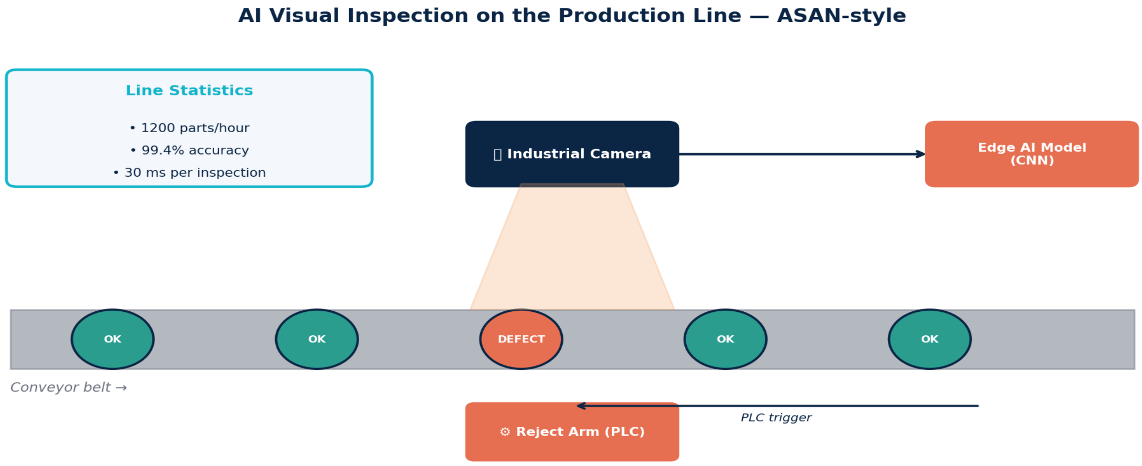

Figure 6.1 — A typical automated visual inspection station. Camera + edge AI + PLC trigger achieves over 99% accuracy at 1000+ parts per hour.

7. Self-Driving Cars & Autonomous Robotics

A self-driving car like a Tesla, Hyundai IONIQ 5 robotaxi, or a Waymo runs a dozen vision models simultaneously — detecting lanes, vehicles, pedestrians, cyclists, traffic signs, and signal states many times per second. Vision is combined with LiDAR (laser distance), radar (long-range distance/speed), GPS and HD maps in a process called **sensor fusion**. The combined view is far more reliable than any single sensor.

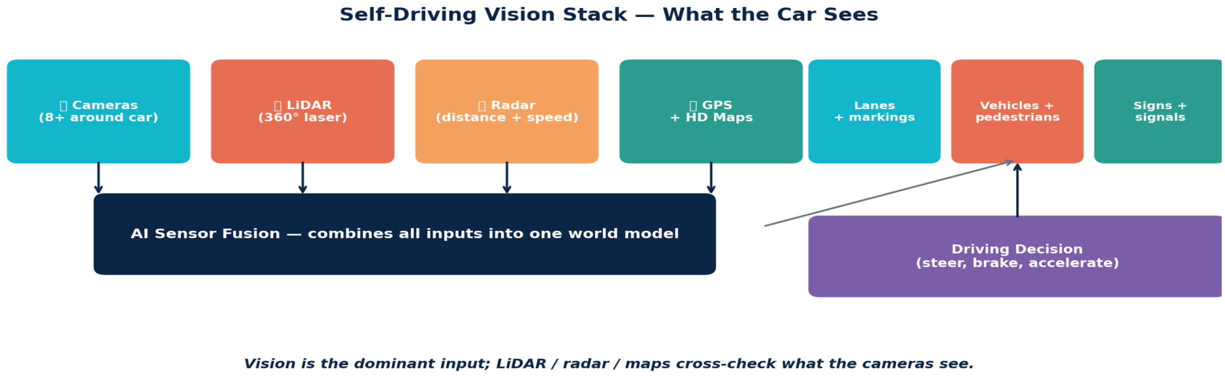

Figure 7.1 — A self-driving vision stack. Multiple sensors feed an AI fusion system that produces a unified world model used for steering, braking, and acceleration decisions.

8. Medical Imaging & Healthcare

AI is transforming radiology. Vision models trained on millions of X-rays, CT scans, MRIs, mammograms and retinal photographs now match — and sometimes exceed — human experts at spotting tumours, fractures, retinopathy, and early-stage cancers. The AI does not replace the doctor; it acts as a tireless second pair of eyes that highlights suspicious areas so the doctor can focus on hard cases and on patient care.

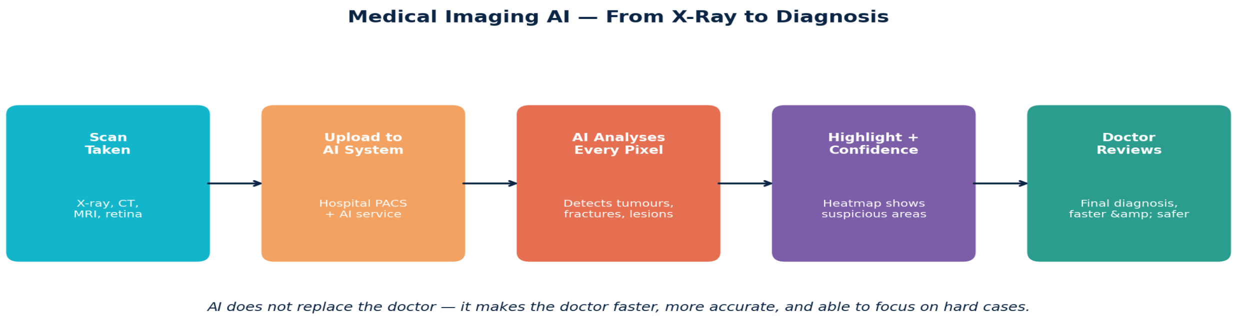

Figure 8.1 — The medical imaging AI workflow. AI highlights what the doctor should look at more carefully — faster reads, fewer missed findings, better outcomes.

9. Edge AI vs Cloud AI

Where should the vision model actually run? On the device itself (called **edge AI**) or on a remote server (**cloud AI**)? Both are widely used, and the right choice depends on speed, privacy, cost, and network availability. A factory inspection station needs millisecond response and works inside a private network — perfect for edge. A smartphone app that analyses your travel photos can wait a second and benefits from the cloud's massive compute power.

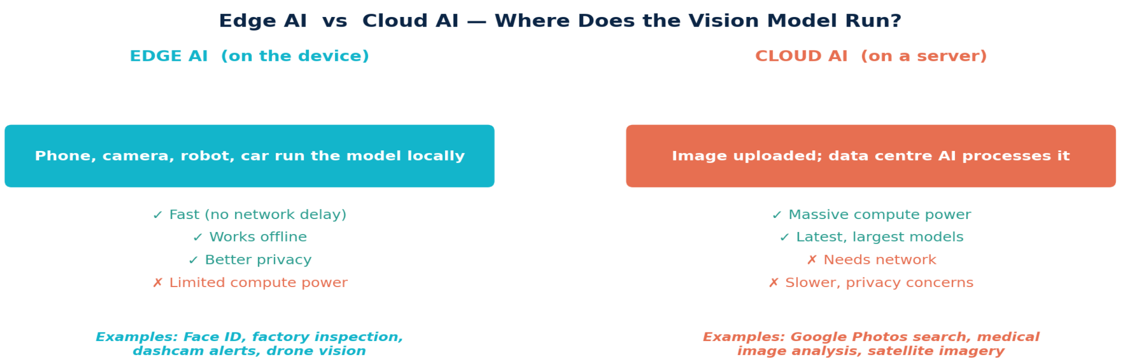

Figure 9.1 — Edge AI vs Cloud AI. The first decision in any production deployment: where does the model live, and what trade-offs does that create?

Criterion	Edge AI	Cloud AI
Speed	Milliseconds (no network)	100 ms+ network round-trip
Privacy	Data stays on device	Data leaves the device
Cost	One-time hardware purchase	Pay per query (often forever)
Model size	Small, optimised	Largest models possible
Offline use	Yes	No — needs internet
Best for	Industrial inspection, robotics, cars, face unlock	Search, large-scale analytics, AI assistants

10. Summary, Self-Check Quiz & Homework

Lesson Recap

• Every vision system runs the same six-stage pipeline: capture → preprocess → extract features → classify → post-process → decide.
• The four core tasks: **classification** (label whole image), **detection** (bounding boxes), **segmentation** (pixel masks), **pose** (keypoints).
• **CNNs** build a feature hierarchy: edges → textures → shapes → objects. They are the default for most engineering vision work.
• **Vision Transformers (ViT)** use attention instead of convolution; they win at very large scale and power multimodal AI.
• **Factory visual inspection** = camera + edge CNN + PLC, 99%+ accuracy at thousands of parts per hour.
• **Self-driving cars** fuse cameras, LiDAR, radar, and HD maps into a single world model.
• **Medical AI** highlights suspicious areas in scans; it supports doctors rather than replacing them.
• **Edge AI** = on the device (fast, private, offline). **Cloud AI** = on a server (largest models, but slower and online).

Self-Check Quiz (answers at the bottom)

1. Name the six stages of the computer vision pipeline.
2. A camera is monitoring a conveyor belt and you only need to know 'OK or defective' for each part. Which vision task should you use?
3. What is the key idea of self-attention in a Vision Transformer?
4. Give two reasons to run a vision model on the edge instead of the cloud.
5. What is sensor fusion, and why is it used in self-driving cars?

Answers: 1) Capture → Preprocess → Extract features → Classify → Post-process → Decide/act. 2) Classification — simplest task, smallest dataset, fastest to deploy. 3) Split the image into patches, treat each as a token, let every patch attend to every other patch via attention. 4) Lower latency (no network round-trip), better privacy (data stays on device), works offline, lower per-query cost. 5) Combining multiple sensor inputs (camera, LiDAR, radar, GPS) into one unified world model — far more reliable than any single sensor alone, especially in bad weather or unusual conditions.

Homework Assignment

Pick a real-world vision problem you would like to solve — defect inspection at your workplace, counting cars in a parking lot, monitoring crop health, reading meter displays, or recognising livestock — and complete the following:

1. Identify which **task type** (classification, detection, segmentation, pose) fits best, and explain why.
2. Where will the camera be placed? Describe the **capture** setup: lighting, distance, angle, frequency.
3. Will the model run on the **edge or in the cloud**? Justify the trade-off.
4. List **three preprocessing steps** you would apply before the AI model.
5. Describe one **failure case** the system might face in the real world, and how you would handle it.

Submit to **awetgknway@gmail.com** with subject *AI Day 8 Homework — Your Name*.

Coming Up Tomorrow — Day 9

Speech, Audio & Voice AI — how AI listens, transcribes, translates, and speaks. You'll learn how Siri and Bixby actually work, how voice cloning is done, and the engineering applications of audio AI in factories (anomaly detection from machine sounds).

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